

Divergence

setup

$$w=4, h=3$$

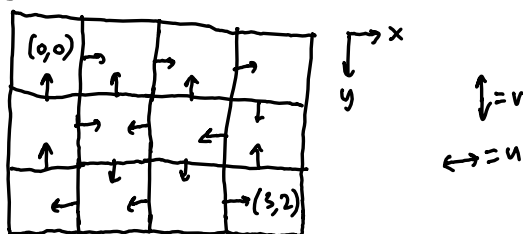
$$\text{dens} = 4 \times 3 = w \times h \rightarrow i + w \cdot j$$

staggered components

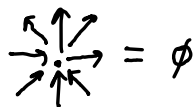
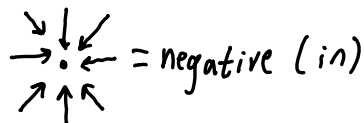
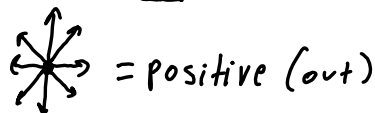
$$u = 3 \times 3 = (w-1) \times h \rightarrow i + (w-1) \cdot j$$

$$v = 4 \times 2 = w \times (h-1) \rightarrow i + w \cdot j$$

grid



conventions



divergence = out flow - in flow

$$\text{div}_{0,0} = u[0,0] + v[0,0] \leftarrow \text{top left corner}$$

$$\text{div}_{1,0} = -u[0,0] + u[1,0] + v[1,0] \leftarrow \text{top edge}$$

$$\text{div}_{0,1} = u[0,1] - v[0,0] + v[0,1] \leftarrow \text{left edge}$$

$$\text{div}_{1,1} = -u[0,1] + u[1,1] - v[1,0] + v[1,1] \leftarrow \text{center}$$

↑ = same

$$\text{div}_{3,2} = -u[2,2] - v[3,1]$$

solve incompressibility

add/sub $\frac{\text{div}}{\#}$ to each component s.t. $\text{div} \rightarrow \emptyset$